

2 TREES / THE BOOK

2 trees: Build your (D)Home

Working manuscript

Illustration placement QA fixture

A reading edition of the current editable manuscript. Illustrations retain their captions and source notes.

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Choose a sphere radius R , then keep one unit throughout the calculation. If the radius is in feet and the working dimensions will be inches, multiply it by twelve before finding the chords.



Figure 1. Concept artwork — workbench-v3. Geometry review required before technical use.

Source: AI-generated editorial artwork; not a build photograph or cut diagram.

The nominal relationships are $A = R \times f_A$ and $B = R \times f_B$. For this project's exact generated 2V mesh, f_A is approximately 0.618033989 and f_B is approximately 0.546533058. The Sizing lab obtains these factors from the existing geometry module rather than keeping the rounded printed values as its authority.

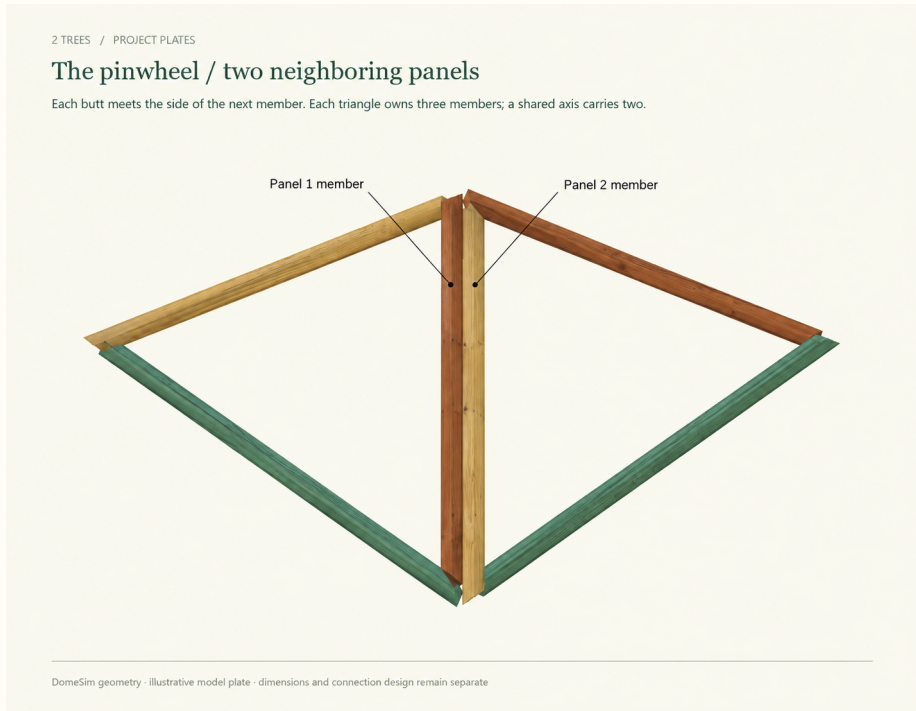


Figure 2. Each central member belongs to a different triangle. Around each triangle, the butt of one member meets the side near the front of the next.

Source: AI-finished illustration using the actual simulator pinwheel-pair render as the geometry reference; original model plate retained in Chapter 17.

The procedure is sequential. Record the desired radius. Generate both chord lengths. Confirm the forty-panel inventory. Select the member cross-section, orientation, and seam construction. Generate the physical model and inspect its actual stock lengths. Only then select raw blank lengths and allocate trees.



Figure 3. The starting inventory is a standing pine. A still from the existing harvest lesson, captured at 30 percent of its opening tree scene; the on-screen number illustrates the source lesson, not a measured harvest result.

Source: Rendered from Two Trees, All at Once using the existing scene painter and authored camera. Source-model example.

If a radius gives nominal edges that are inconvenient for the timber, the design still has options. A slightly smaller radius can reduce length demand. A different seam or corner setting can change the stock envelope. More usable source length may be available in a longer section. Each option must be checked through the same solver; none should be hidden inside an unexplained rounding choice.



Figure 4. A snapshot of the existing felling animation. The scene depicts the change from standing tree to felled trunk; it is a visual explanation from the source film.

Source: Rendered from Two Trees, All at Once / fell at 62%, using the original video scene and camera (original overlay). Source model assumptions, not verified field results. Timing uses authored chapter durations; source hashes and the complete snapshot recipe are saved with this image.

Record three lengths for each member: the ideal chord, the finished physical stock envelope, and the blank required to fabricate it. The last includes whatever additional holding, trimming, and overfit stock the actual process needs. Zero extra allowance in the lab explicitly means that only the finished envelope is being checked.



Figure 5. Eight six-foot sections, eight potential wedges per section: 64 gross blanks per tree and 128 from two trees. The 120-member frame target leaves eight gross blanks before real defects and losses are counted.

Source: Rendered from Two Trees, All at Once / explode at 88%, using the original video scene and camera (original overlay). Source model assumptions, not verified field results. Timing uses authored chapter durations; source hashes and the complete snapshot recipe are saved with this image.

When the stock plan is complete, export the actual simulator's bill of materials and cut schedule for the chosen configuration. The book can explain the reasoning, while that schedule preserves the geometry used for the build. Write the configuration and date beside any printed schedule so an older paper does not silently govern a newer design.

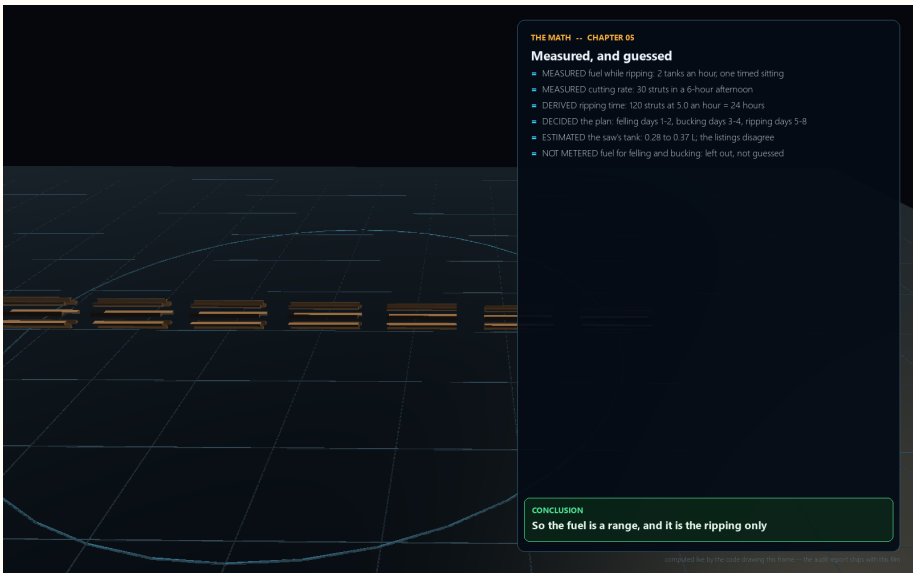


Figure 6. The source lesson's math overlay preserves measured, derived, decided, estimated and unmetered labels. This is a snapshot of its current example; the source values are printed above for reading at book size.

Source: Rendered from Two Trees, All at Once / assume at 95%, using the original video scene and camera (math overlay). Source model assumptions, not verified field results. Timing uses authored chapter durations; source hashes and the complete snapshot recipe are saved with this image.